

Transatlantic clean investment monitor 4: electric vehicles

A clear move to electric vehicles is underway, but the US is creating more obstacles than Europe

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This is the fourth of a series of Bruegel-Rhodium Group quarterly briefings to compare clean tech deployment and manufacturing trends in Europe and the United States.

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*Click [here](#) to access the **European clean tech tracker** dataset*

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Electric vehicles (EVs) have grown from a niche technology to a mainstream industry in the space of a little over a decade. EV deployment is crucial to meet climate targets and is transforming the automotive industry – a cornerstone of manufacturing in both the United States and Europe. Unlike China, the EU and US are particularly exposed to the EV transition with whole communities built on the back of internal combustion engine (ICE) technology.

The roll-out of EVs is driven by policy incentives, falling costs and shifting consumer preferences. To meet growing demand in the EU and US, companies are investing hugely across the EV value chain. Fresh competition comes from booming Chinese manufacturing capacity, which policymakers and companies are grappling with.

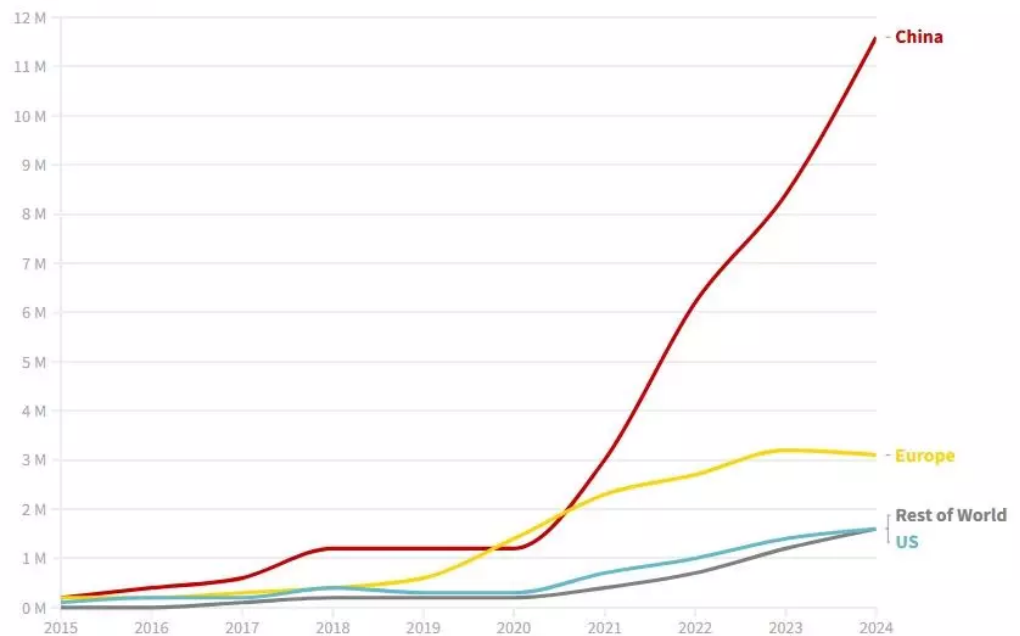
Based on an assessment of EV uptake in major regions – one in ten new vehicles in the US is an EV, compared to one in five in the EU, and one in two in China – we evaluate EV manufacturing investment in the EU and US at the project level, to provide a transatlantic perspective on the scale and pace of regional investment and what that means for EV production capacities. The data helps assess how invested the US and EU are in producing electric vehicles, and how exposed each is to Chinese competition.

EV sales

Global demand for EVs has accelerated since 2020. In China, sales and investment across the domestic EV supply chain took off in the late 2010s, driven by incentives for manufacturers and consumers. China now dominates the global EV market, with 65 percent of global sales in 2024, followed by Europe¹ with 20 percent and the US with 10 percent.

European sales rose in 2019 after the EU passed legislation to ban the sale of new ICE vehicles by 2035, introducing annual fleet-wide CO₂ emission targets for all new vehicles (Figure 1). Under pressure from auto manufacturers, in May 2025, the EU requirements were modified. Rather than requiring companies to meet annual CO₂ emissions targets for new cars in 2025, 2026 and 2027 separately, the rules now allow compliance to be assessed based on the average over those three years. While this somewhat weakens ambition and creates investment uncertainty, the 2035 target remains. Norway and the UK have similar phase-out plans with zero new-ICE sales from 2025 and 2030, respectively.

In the US, federal tailpipe standards and EV tax credits, coupled with state-level EV mandates, helped drive EV take-up in the late 2010s. The 2022 Inflation Reduction Act (IRA) expanded consumer EV tax credits.

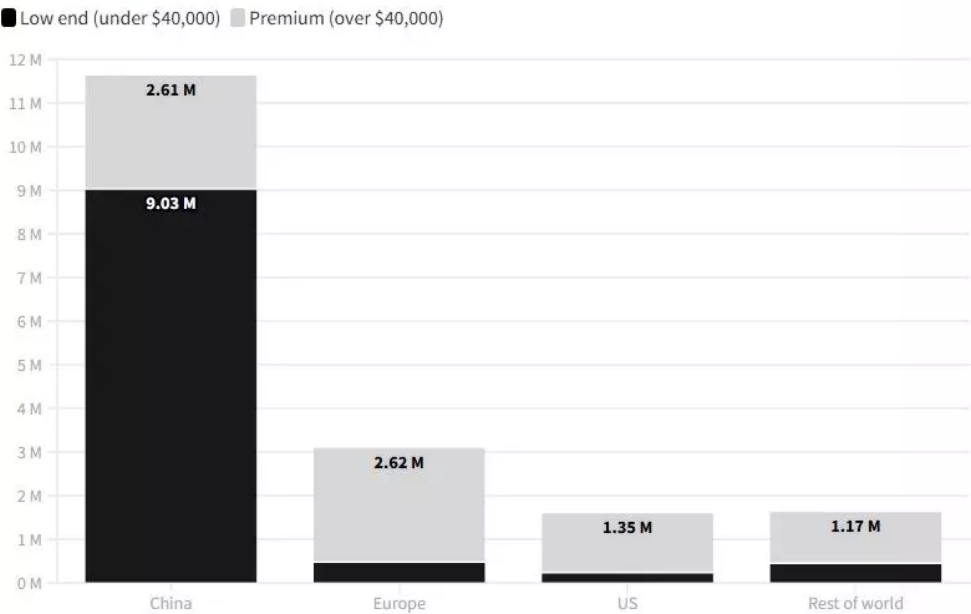
Figure 1: EV sales by region, 2015-2024, millions.

Source: Bruegel and Rhodium based on EV Volumes.



Sales in China are dominated by 'low-end' vehicles priced under \$40,000, while sales in Europe and the US are more concentrated in the premium segment (Figure 2). The average price of a Chinese EV is \$35,100, compared to \$56,800 in Europe and \$59,100 in the US². Faced with limited EV models at an affordable price, many European and US consumers have continued to opt for conventional vehicles. Consumer preferences, coupled with the phase-out of subsidies in EU countries and policy uncertainty in the US, led to slowing sales growth in both regions in 2024.

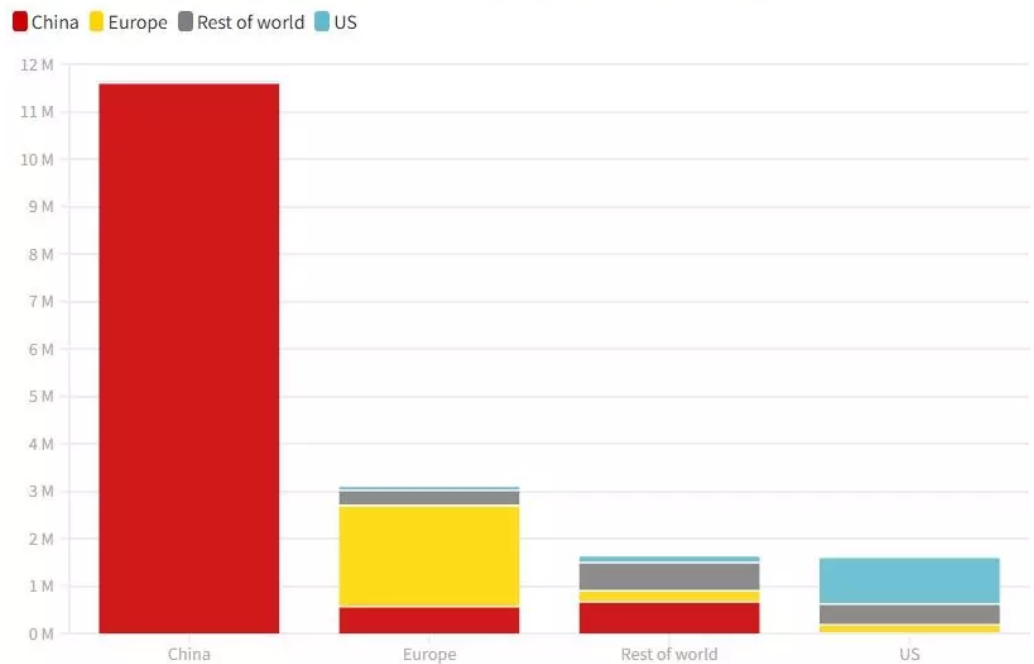
Figure 2: EV sales by market segment , China, Europe, US and rest of the world, millions, 2024



Source: Bruegel and Rhodium based on EV Volumes. • Note: Low end indicates vehicles sold for less than \$40,000, while premium indicates vehicles sold for more than \$40,000.



Two-thirds of EVs sold in Europe and the US are produced locally – with local production referring to final assembly of vehicles, not earlier stages such as battery cell fabrication (Figure 3). Chinese-produced EVs are one-fifth of Europe’s market, with essentially no sales in the US. Domestically, China builds nearly everything it sells. In the rest of the world, where EV sales are rising quickly, China accounts for 40 percent of sales, with the US and Europe taking smaller slices.

Figure 3: EV sales by country of origin in 2024, millions

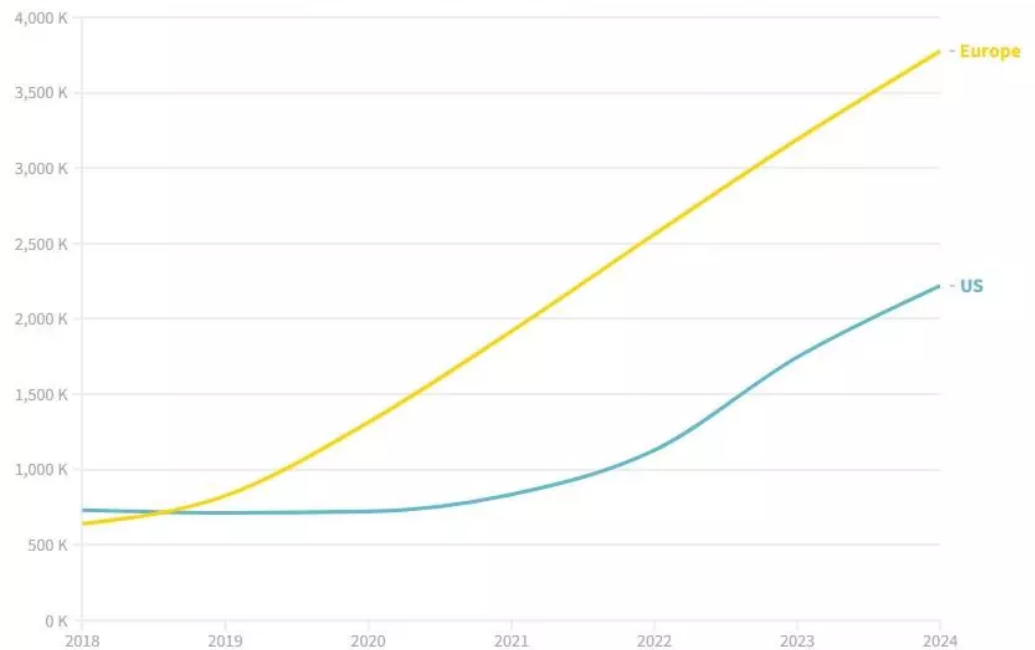
Source: Bruegel and Rhodium based on EV Volumes.



EV manufacturing investments and production capacity

The US and Europe are global leaders in ICE manufacturing. Consequentially, the auto industries in the US and Europe have developed strong political constituencies relatively insulated from overseas competition, giving US and European automakers weaker incentives to invest in novel EV technologies, compared to Chinese producers.

As global EV sales rise and concerns about China's leadership in the sector grow, both Europe and the US have ramped up domestic manufacturing. In Europe, growing demand has driven a surge in EV production capacity since the CO₂ standards were passed in 2019, often supported by national state aid (Figure 4). The US saw an uptick in EV production capacity following the adoption of the IRA, but capacity is around two-thirds of that in Europe.

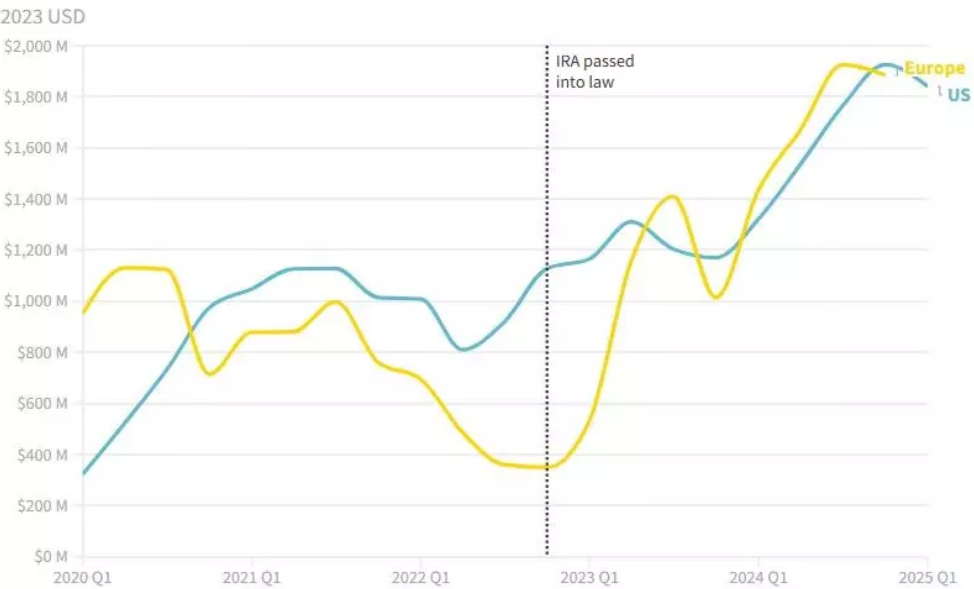
Figure 4: EV production capacity, Europe and US, thousands

Source: Bruegel European Clean Tech Tracker and Rhodium Group-MIT CEEPR Clean Investment Monitor.



Annual investment has grown steadily in both regions, at roughly 10 percent quarter-on-quarter from 2020 to 2025 (Figure 5). Even with this impressive growth, compared to batteries, the EV industry in the US saw less of a step change in investment following the passage of the IRA compared to batteries (Jugé *et al*, 2025). Investment in the battery industry surged after the IRA, and the industry diverged sharply from Europe (Figure 6). Many factors may contribute to the more modest impact of the IRA on EVs relative to batteries, including the fact that IRA tax credits apply to a wide array of battery components (including cells, modules, packs, anodes and cathodes), leaving room for a variety of new entrants. By contrast, EVs are a more complex product and therefore new ventures likely face higher barriers to entry and higher bars to meet domestic sourcing requirements for IRA tax credits.

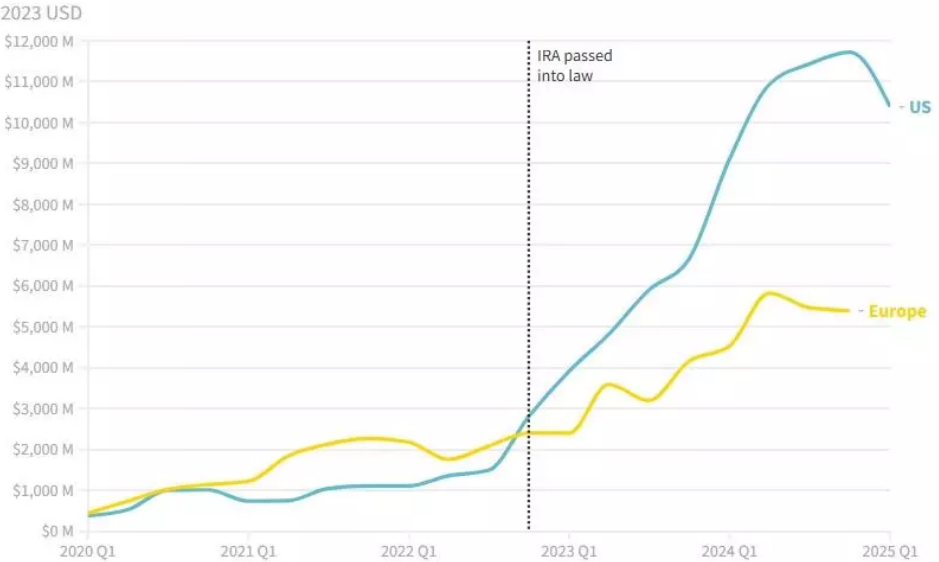
Figure 5: Quarterly real investments in EV production, Europe and US, \$ millions



Source: Bruegel European Clean Tech Tracker and Rhodium Group-MIT CEEPR Clean Investment Monitor. • Note: data covers operational and planned investment through Q1 2025 for the US and Q4 2024 for Europe



Figure 6: Quarterly real investments in the battery manufacturing supply chain, Europe and US, \$ millions



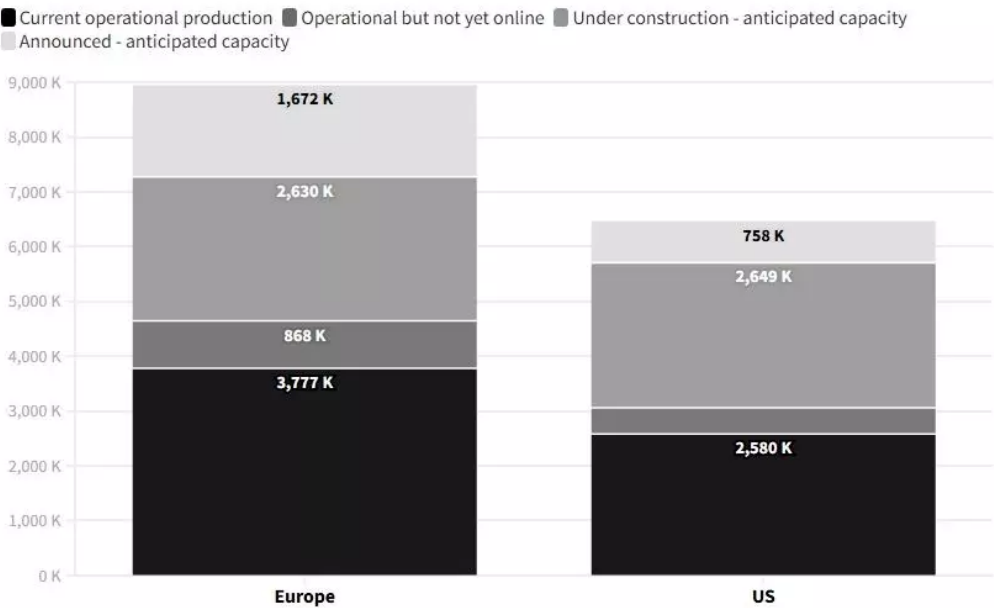
Source: Bruegel European Clean Tech Tracker and Rhodium Group-MIT CEEPR Clean Investment Monitor. • Note: data covers operational and planned investment through Q1 2025 for the US and Q4 2024 for Europe



There are signs of a slowdown in EV investment growth rates in both markets, even though investments remain high compared to earlier, and companies continue to make new announcements. In the US, growth in EV manufacturing investments decelerated in Q4 2024, before falling 5 percent in Q1 2025. In addition, more than \$6 billion worth of battery projects, including Freyr Battery in Georgia and Kore Power in Arizona, were cancelled in Q1 2025³ – the highest value of cancellations on record. Quarter-on-quarter European investment also slowed in Q3 2024 and shrank marginally in Q4 2024. In Europe, ACC and SVOLT have paused their plans for battery manufacturing plants in Germany⁴. In both regions, this downturn mirrors slowing EV sales. In the US, it is compounded by uncertainty about the future of federal EV support and the impact of tariffs.

These uncertainties create additional risks for the pipeline of future projects, including operating facilities ramping up production ('operational but not yet online', Figure 7) and those under construction and announced. For the US, a capacity of 3.9 million vehicles is anticipated to come online, adding to current operational capacity of 2.6 million. Europe has even more new projects in the pipeline, with capacity for 5.2 million vehicles under construction or announced, in addition to 3.8 million in capacity already available (Figure 7). While the outlook appears more stable for Europe, the rate of future EV demand growth remains uncertain.

Figure 7: EV Production Capacity by project status, Europe and US, thousands



Source: Bruegel European Clean Tech Tracker and Rhodium Group-MIT CEEPR Clean Investment Monitor. • *Note: data covers operational and planned investments through Q1 2025 for the US and Q4 2024 for Europe.



The future competitiveness of the EV sector

President Trump's tariffs, the risk of escalating trade wars and an evolving policy landscape will impact EV sales and production in both the US and Europe. In the US, EVs have become a political flashpoint and the industry's future is uncertain. Even before Trump's second term, EV prices were relatively high in the US. Tariffs on construction materials will make it more expensive to build manufacturing facilities. Tariffs on China – initiated under President Biden and ramped up under President Trump – raise the cost of EV production in existing facilities, given that China provides most upstream EV battery components.

The budget package passed by the US House of Representatives on 22 May 2025, and at time of writing under consideration in the Senate, would effectively gut the energy tax credits under the IRA, including the EV-related provisions that have been critical in catalysing the latest wave of investments⁵. Potential repeal of the IRA, along with many other attacks on federal⁶ and state clean vehicle regulation and incentives⁷, further undermine the outlook for the US domestic market and threaten to reduce the competitiveness of US-made EVs abroad.

The EU outlook is clearer, with the 2035 zero-emission vehicles target providing certainty. The main question is the degree to which domestic production should be protected, even at the cost of restricting cheaper imports, which benefit consumers. This issue has arisen in discussions concerning whether the EU should impose tariffs on imports of Chinese-made electric vehicles⁸. Imports of Chinese-made EVs are important for increasing the availability of affordable models to European consumers. But taking market share from European production is damaging for domestic jobs and value added, and especially a concern if deemed unfair competition due to Chinese state subsidies.

In response, the EU launched an anti-subsidy investigation in 2023 and by October 2024 imposed additional tariffs⁹ on top of the standard 10 percent import duty. These tariffs aim to address unfair competition while still allowing Chinese EV imports to continue. Currently, the EU and China are negotiating the future of these trade measures. One option under discussion is the introduction of a minimum import price mechanism, which would establish a floor price for Chinese EVs sold in Europe¹⁰.

While the US and European EV industries are primarily focused on domestic markets, both exported just under one-fifth of their domestic EV production in 2024. European manufacturers have diversified their export partners and fostered partnerships with Chinese firms, a strategy their US counterparts have largely avoided.

The global EV transition presents an enormous industrial opportunity. China's significant lead in production scale, cost efficiency and domestic market size positions it as the dominant player. US and European firms are likely to lose the export shares they enjoyed during the ICE era, both to the Chinese market and other countries. The extent to which Chinese exporters are allowed into the US and European markets will continue to be an important policy question, with a need to balance affordability and industrial competitiveness. Current US policy is to deny any domestic market access to Chinese exporters, while the picture in Europe is more nuanced. Smart policy is needed to strengthen domestic markets, encourage domestic manufacturing and respond to growing Chinese dominance.

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Endnotes

- 1 The EU, Norway, Switzerland and the United Kingdom.
- 2 Sales-weighted average across all car types (including SUVs). Calculated using data from EV Volumes.
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- 6 For example, the Environmental Protection Agency announced plans to reconsider and potentially roll back emissions standards for light-duty and medium-duty vehicles for model years 2027 and later. See US EPA press release of 12 March 2025, 'EPA Announces Action to Implement POTUS's Termination of Biden-Harris Electric Vehicle Mandate', <https://www.epa.gov/newsreleases/epa-announces-action-implement-potuss-...>
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